Exact calibration tests for strain, pressure and angular spectral ratios

Project Stardust

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Abstract

We assemble exact diagnostics for velocity gradients, spectral ratios and threshold occupation. A family of nonnormal restricted Euler matrices has fixed full spectrum but variable strain spectrum, with arbitrarily small positive normalized stretching along trajectories whose raw stretching diverges. Its affine realization solves the local Navier–Stokes equations with quadratic pressure, but has infinite whole-space energy. A trace computation then detects an incompatible mixture of full and trace-free pressure conventions: the proposed strain right side has trace equal to the squared Frobenius norm of the velocity gradient under the pressure Poisson constraint. Finally a smooth periodic viscous ABC flow has zero averaged stretching derivative, while a displayed plus-pressure balance gives three at the initial time. A fixed smooth periodic Gaussian-series shear has a specified normalized two-sector spectral ratio strictly above one at every finite nonnegative time, with its full lattice spectrum and antipodal conic partition retained. For threshold labels, an exact null-boundary criterion separates stable occupation from smooth positive-sector counterexamples. Complete elementary proofs reconstruct frozen formal results. These are calibration examples in classical model families, not new fluid existence or singularity results; priority for the diagnostic formulations remains unresolved. An actual stationary Burgers vortex additionally separates an axial pressure flux from a scalar Gaussian observation, and a repaired helical field gives an exact finite-slab base-pressure pairing.

1 Conventions and purpose

For a real 3×3 matrix A, use

$$S = \tfrac{1}{2}(A + A^{\mathsf{T}}), \qquad W = A - S, \qquad \omega = (A_{32} - A_{23}, A_{13} - A_{31}, A_{21} - A_{12}).$$

For an actual velocity u, $A_{ij} = \partial_j u_i$ and $\omega = \nabla \times u$. The pressure Hessian is $P = \nabla^2 p$, so the incompressible pressure constraint is

$$-\operatorname{tr} P = \operatorname{tr}(A^2). \tag{1}$$

This convention agrees with the velocity-gradient formulation in Cantwell [1] and the explicit pressure equations of Yang, Xu, Pumir and He [2]. Full pressure and its deviatoric part $P_0 = P - \frac{1}{3}\operatorname{tr}(P)I$ must be distinguished.

The first test concerns the matrix ODE

$$A' = -A^2 + \tfrac{1}{3}\operatorname{tr}(A^2)I, \qquad \operatorname{tr} A = 0. \tag{2}$$

The second is an algebraic test of a specified strain right side. The third tests a specified spatially averaged identity on an actual smooth periodic Navier–Stokes flow. Each test has its own hypotheses; neither an ODE trajectory nor an infinite-energy affine field is promoted to a finite-energy singularity theorem.

2 Fixed full spectrum and arbitrarily small normalized stretching

For $p > 2$ set $a = \sqrt{(p-2)(p+1)} > 0$ and define

$$M_{p,a} = \begin{pmatrix} -p & 0 & a \\ 0 & 1 & 0 \\ -a & 0 & p-1 \end{pmatrix}, \qquad c(t) = (1-t)^{-1}, \qquad A(t) = c(t) M_{p,a}. \tag{3}$$

For $\omega \neq 0$ let

$$\sigma(A) = \frac{\omega^{\mathsf{T}} S \omega}{|\omega|^2}, \qquad \rho(A) = \frac{\sigma(A)}{\|S\|_F}.$$

The normalization uses the Frobenius norm of the symmetric strain, not the spectral radius of the nonsymmetric full matrix.

Theorem 2.1. *For every $p > 2$ the trajectory (3) solves (2) for $t < 1$. Its strain eigenvalues in increasing order are $c(t)(-p, 1, p-1)$, and its nonzero vorticity is aligned with the intermediate eigendirection. Its full characteristic polynomial is*

$$\det(rI - A(t)) = (r - c(t))^2 (r + 2c(t)),$$

whereas

$$\sigma(A(t)) = c(t), \qquad \rho(A(t)) = \frac{1}{\sqrt{p^2 + 1 + (p-1)^2}}. \tag{4}$$

For every $\epsilon > 0$ a choice of $p > 2$ makes $0 < \rho(A(t)) < \epsilon$ for all $0 \le t < 1$, while $\sigma(A(t)) \to \infty$ as $t \uparrow 1$. No positive uniform lower threshold for this normalized quantity holds on these exact profiles.

Proof. The middle coordinate of M is one. On the first and third coordinates, direct multiplication and $a^2 = p^2 - p - 2$ give

$$M^2 + M = 2I, \qquad \operatorname{tr} M = 0, \qquad \operatorname{tr} M^2 = 6.$$

Thus $-M^2 + \operatorname{tr}(M^2) I/3 = M$. Since $c' = c^2$ and the ODE right side is homogeneous of degree two, $A' = c^2 M$ is precisely that right side. Also

$$S(M) = \operatorname{diag}(-p, 1, p-1), \qquad \omega(M) = (0, 2a, 0).$$

Multiplying by $c > 0$ proves the strain ordering and alignment and gives $|\omega(A)|^2 = 4a^2 c^2$ and $S(A)\omega(A) = c\omega(A)$. The stretching formulas (4) follow, since $\|S(A)\|_F^2 = c^2(p^2 + 1 + (p-1)^2)$.

The determinant factors as

$$\det(rI - M) = (r-1)((r+p)(r-p+1) + a^2) = (r-1)^2 (r+2).$$

Rescaling gives the stated polynomial for A. The matrix is nonnormal: the $(1,3)$ entries of $M^{\mathsf{T}} M$ and $M M^{\mathsf{T}}$ are $-a(2p-1)$ and $a(2p-1)$, respectively, and are unequal. Hence the varying symmetric spectrum does not contradict the fixed full spectrum.

For $\epsilon > 0$, choose $p = 2 + \epsilon^{-1}$ and the specified a. The denominator in (4) is strictly larger than p, and $\epsilon p = 2\epsilon + 1 > 1$, so $0 < \rho < 1/p < \epsilon$. Meanwhile $c(t) \to \infty$ as $t \uparrow 1$, proving raw stretching divergence along each selected trajectory. □

For example $p = 3, a = 2$ gives strain ratios $-3 : 1 : 2$; $p = 4, a = \sqrt{10}$ gives $-4 : 1 : 3$ and normalized stretching $1/\sqrt{26}$. A particular strain ratio is therefore not universal even within this elementary self-similar family. Cantwell's general restricted-Euler solution already distinguishes full-gradient and strain/rotation geometry [1]. The examples here are exact calibration data, not a new solution theory.

3 An affine realization and its spatial limitation

Proposition 3.1. *With the same parameters, define on $\mathbb{R}^3$, for $t < 1$,*

$$u(t,x) = c(t)M_{p,a}x, \qquad p_{\mathrm{aff}}(t,x) = -c(t)^2|x|^2.$$

These smooth fields satisfy

$$\nabla \cdot u = 0, \qquad \partial_t u + (u\cdot\nabla)u + \nabla p_{\mathrm{aff}} = \nu\Delta u$$

for every fixed $\nu \geq 0$, pointwise on this time interval. They are not periodic and $u(t,\cdot) \notin L^2(\mathbb{R}^3)$.

Proof. The spatial gradient is cM, the divergence is $c\,\mathrm{tr}\,M = 0$, and $\Delta u = 0$. Moreover

$$\partial_t u = c^2 Mx, \qquad (u\cdot\nabla)u = c^2M^2x, \qquad \nabla p_{\mathrm{aff}} = -2c^2x.$$

The momentum identity is therefore $c^2(M + M^2 - 2I)x = 0$. Its pressure Hessian is $P = -2c^2I$ and satisfies $-\mathrm{tr}\,P = 6c^2 = \mathrm{tr}(A^2)$, so the Poisson convention is consistent. The middle velocity coordinate is $u_2 = cx_2$, which is not periodic. Since $c > 0$, integrating $|u|^2 \geq c^2x_2^2$ over expanding boxes gives an infinite integral. Thus these fields do not meet the whole-space finite-energy hypothesis. □

They also provide a local pressure-sign test. Define $q = \omega^{\mathsf{T}}S\omega$, $b = |S\omega|^2$ and $h = \omega^{\mathsf{T}}P\omega$ for this affine field. Directly,

$$q = 4a^2c^3, \qquad b = 4a^2c^4, \qquad h = -8a^2c^4.$$

Since q is spatially constant and $c' = c^2$,

$$(\partial_t + u\cdot\nabla)q = 12a^2c^4 = b - h \neq b + h.$$

This equality is a computation for these fields, not a proposed general formula with all other terms suppressed. It calibrates the sign of the displayed pressure contraction locally; it supplies no finite global integrals of the affine field.

4 A trace obstruction to mixed pressure conventions

The strain right side to be tested is the explicit matrix expression

$$R_+(A,P) = -S^2 - W^2 - P + \frac{\mathrm{tr}(S^2) + |\omega|^2/2}{3}I. \tag{5}$$

It is this expression, including the isotropic plus sign and the meaning of P, that the next result excludes as a trace-preserving incompressible strain evolution.

Theorem 4.1. *For every real A, P satisfying* (1)*,*

$$\mathrm{tr}\,R_+(A,P) = \|A\|_F^2. \tag{6}$$

In particular it is strictly positive if $A \neq 0$. Adding a trace-free matrix correction cannot restore trace zero. The full-pressure expression and the trace-free-pressure expression

$$R_{\mathrm{full}}(A,P) = -S^2 - W^2 - P, \tag{7}$$

$$R_{\mathrm{dev}}(A,P_0) = -S^2 - W^2 - P_0 + \frac{\mathrm{tr}(S^2) - |\omega|^2/2}{3}I \tag{8}$$

are trace-consistent under their respective conventions. Under (1)*,* (7) *and* (8) *agree.*

Proof. The symmetric/skew splitting is Frobenius orthogonal. The skew matrix associated with the above vorticity has squared Frobenius norm $|\omega|^2/2$, and hence

$$\begin{aligned}\operatorname{tr} W^2 &= -|\omega|^2/2,\\ \operatorname{tr} A^2 &= \operatorname{tr} S^2 - |\omega|^2/2,\\ \|A\|_F^2 &= \operatorname{tr} S^2 + |\omega|^2/2 = \operatorname{tr} A^2 + |\omega|^2.\end{aligned}$$

For example $|\omega|^2$ is the sum over $i<j$ of $(A_{ij}-A_{ji})^2$, while the off-diagonal contribution to $\operatorname{tr} A^2$ is $2\sum_{i<j} A_{ij}A_{ji}$; adding proves the last identity directly. Taking the trace of (5) gives $|\omega|^2 - \operatorname{tr} P$. Substituting (1) proves (6). The Frobenius squared norm is positive unless all entries vanish, and a trace-free addition leaves this trace unchanged.

The trace of (7) is $-\operatorname{tr} A^2 - \operatorname{tr} P$, which is zero under (1). The trace of (8) is zero because $\operatorname{tr} P_0 = 0$ and $\operatorname{tr}(S^2) - |\omega|^2/2 = \operatorname{tr} A^2$. Finally $P = P_0 + \operatorname{tr}(P)I/3 = P_0 - \operatorname{tr}(A^2)I/3$ under the constraint; inserting this in (7) gives (8). □

If $\operatorname{tr} A = 0$, the strain trace must remain zero along an incompressible evolution. Theorem 4.1 thus gives an immediate nonzero-state obstruction to (5). The theorem is a consistency test, not a derivation or closure of a nonlocal pressure law.

5 A periodic positive-viscosity pressure-sign test

On the 2π-periodic three-torus let

$$U(x,y,z) = (\sin z + \cos y,\ \sin x + \cos z,\ \sin y + \cos x).$$

This is the classical unit-parameter ABC profile; the same convention is used in [3]. Write $\langle f\rangle = (2\pi)^{-3}\int_{[0,2\pi]^3} f$ for normalized spatial average. For a smooth velocity v with strain S_v and vorticity ω_v, define

$$Q(v) = \langle \omega_v^{\mathsf{T}} S_v \omega_v\rangle, \quad B(v) = \langle |S_v\omega_v|^2\rangle, \quad H(v,p) = \langle \omega_v^{\mathsf{T}}(\nabla^2 p)\omega_v\rangle,$$

and the specific viscous contraction

$$J(v) = \left\langle \sum_{k,i,j} (S_v)_{ij}(\partial_k\omega_{v,i})(\partial_k\omega_{v,j}) \right\rangle.$$

These are ordinary derivatives and integrals of smooth periodic fields.

Theorem 5.1. *For each $\nu > 0$, the fields*

$$u(t) = e^{-\nu t}U, \qquad p(t) = -\tfrac{1}{2}|u(t)|^2$$

are a smooth periodic Navier–Stokes solution for all $t \geq 0$. They satisfy $Q(u(t)) = 0$ for every t, whereas at $t = 0$,

$$B(U) = \tfrac{3}{2}, \qquad H(U, -|U|^2/2) = \tfrac{3}{2}, \qquad J(U) = 0. \tag{9}$$

In particular the identity

$$\frac{d}{dt}Q(u(t)) = B(u(t)) + H(u(t),p(t)) - 2\nu J(u(t)) \tag{10}$$

fails at $t = 0$: its sides are zero and three respectively. Reversing the pressure sign makes the two sides agree on this test at $t = 0$, without asserting a general evolution formula from this agreement.

Proof. Differentiation gives $\nabla \cdot U = 0$, $\nabla \times U = U$ and $\Delta U = -U$. For a Beltrami field, $(U \cdot \nabla)U = \nabla K$, $K = |U|^2/2$, by the vector identity $(U \cdot \nabla)U = \nabla K - U \times (\nabla \times U)$. Moreover $S_U U = \nabla K$, because the skew part of ∇U annihilates its vorticity U. For $u = e^{-\nu t}U$, the nonlinear advection is canceled by ∇p, and $\partial_t u = -\nu u = \nu \Delta u$. This proves the PDE assertion.

Periodic integration by parts and $\nabla \cdot U = 0$ give $Q(U) = \langle U \cdot \nabla K \rangle = 0$. At time t, stretching is scaled by $e^{-3\nu t}$, so $Q(u(t)) = 0$ and its derivative is zero. The kinetic gradient is

$$\nabla K = (\cos x \cos z - \sin x \sin y,\ \cos y \cos x - \sin y \sin z,\ \cos z \cos y - \sin z \sin x).$$

Each component has mean square $1/4 + 1/4 = 1/2$; the cross terms have zero average by separation of variables. Since $S_U U = \nabla K$, this proves $B(U) = 3/2$. For the pressure contraction, integrating $U_i U_j \partial_i \partial_j (-K)$ by parts in i gives

$$H(U, -K) = -\langle ((U \cdot \nabla)U) \cdot \nabla(-K) \rangle = \langle |\nabla K|^2 \rangle = \tfrac{3}{2}.$$

All boundary terms cancel by periodicity.

It remains to check the precise viscous contraction, rather than omit it. The strain has zero diagonal and

$$S_{12} = \tfrac{1}{2}(\cos x - \sin y), \quad S_{13} = \tfrac{1}{2}(\cos z - \sin x), \quad S_{23} = \tfrac{1}{2}(\cos y - \sin z).$$

As $\omega_U = U$, the three terms $(\partial_k U)^{\mathsf{T}} S_U (\partial_k U)$, for $k = x, y, z$, are

$$\begin{aligned} &-(\cos y - \sin z)\cos x \sin x, \\ &-(\cos z - \sin x)\sin y \cos y, \\ &-(\cos x - \sin y)\cos z \sin z. \end{aligned}$$

Each has zero normalized average; for example integration in x kills the first. Thus $J(U) = 0$. These values prove both the failure of (10) and the limited sign-reversal calibration. □

The affine sign test and periodic test address different possible loopholes. The former is local and has no finite whole-space energy; the latter is smooth, periodic, and has arbitrary positive viscosity. Neither asserts that all pressure contractions have one sign, or that changing one sign establishes a complete general strain balance.

6 A fixed smooth infinite-spectrum shear and an angular ratio

The preceding periodic example tests pressure signs. We now test a different, explicitly defined spectral ratio, using a second actual periodic solution. The heat-governed shear mechanism is classical; the purpose is to retain the complete spectrum, all denominator signs and the angular partition when evaluating the formula.

6.1 The actual solution and every Fourier coefficient

On the 2π-periodic three-torus with coordinates (x_0, x_1, x_2), put

$$\Theta_s(y) = \sum_{n \in \mathbb{Z}} e^{-sn^2} \cos(ny), \qquad s > 0,$$

and for $\nu \geq 0, t \geq 0$ set

$$u_\nu(t, x) = \left(2\Theta_{1+\nu t}(x_1) + e^{-\nu t} \sin x_2,\ 0,\ 0\right), \qquad p = 0. \tag{11}$$

The initial smoothing is fixed at one; it is not changed as a cutoff or observation horizon varies.

Lemma 6.1 (Classical periodic shear)**.** *Formula* (11) *is a smooth spatially periodic incompressible Navier–Stokes solution for every nonnegative viscosity, with the printed zero pressure. Its normalized integral-defined Fourier coefficients have only the following nonzero first components:*

$$\widehat{u}_0(t,(0,n,0)) = 2e^{-(1+\nu t)n^2} \quad (n \in \mathbb{Z}), \tag{12}$$

$$\widehat{u}_0(t,(0,0,1)) = -\tfrac{i}{2}e^{-\nu t}, \qquad \widehat{u}_0(t,(0,0,-1)) = \tfrac{i}{2}e^{-\nu t}. \tag{13}$$

All other components and frequencies vanish. In particular the zero coefficient is two, and the solution has nonzero modes beyond every finite axis cutoff at every finite time.

Proof. For every integer $m \geq 0$ and $s_0 > 0$, the series $\sum_n |n|^m e^{-s_0 n^2}$ converges. For large $|n|$ its polynomial factor is bounded by $e^{s_0 n^2/2}$, and the resulting Gaussian tail is bounded by a convergent geometric tail. Thus every spatial derivative of the series converges uniformly for $s \geq s_0$. Parameter derivatives introduce additional powers of n^2 and obey the same bound. This justifies repeated differentiation and gives $\partial_s \Theta_s = \partial_y^2 \Theta_s$.

The first component of u is independent of x_0, and the other components vanish. Hence its divergence and its advection $(u \cdot \nabla)u$ both vanish. The theta heat identity and $\partial_t(e^{-\nu t}\sin x_2) = \nu\partial_{x_2}^2(e^{-\nu t}\sin x_2)$ give $\partial_t u = \nu \Delta u$. This proves the complete momentum equation with zero pressure. Smoothness at each nonnegative time follows from $1 + \nu t \geq 1$ and the uniform derivative estimates; the formulas are defined on an open neighborhood of each such time with positive smoothing parameter. Periodicity follows termwise.
Define $\widehat{u}(k)$ by normalized integration against $e^{-ik\cdot x}$. Uniform absolute convergence allows interchanging the theta sum with all three finite-interval integrals. Pairing $n, -n$ gives $\Theta_s(y) = \sum_n e^{-sn^2} e^{iny}$, since the imaginary parts cancel. Orthogonality of the exponentials yields (12); the elementary Fourier expansion of sine yields (13). Integration in the unused coordinate kills all remaining frequencies. Every displayed axis weight is strictly positive at a finite time, which proves the tail claim. □

The original declarations are in
`GaussianFourierJets`, `GaussianPeriodicHeat`,
`GaussianInfiniteShear`, `GaussianShearSpectrum` and `GaussianShearFourierTail`. This is a real smooth finite-energy solution on the torus. It is not the infinite-energy affine construction used earlier.

6.2 Actual angular sectors and exact complete moments

Partition the entire lattice $\mathbb{Z}^3$ into

$$\mathcal{C}_0 = \{k : |k_2| \leq |k_1|\}, \qquad \mathcal{C}_1 = \{k : |k_2| > |k_1|\}.$$

This partition is antipodal and invariant under positive integer scaling. It is an explicitly specified conic partition; no identification with a different manuscript tessellation is assumed. Define, for $j = 0, 1$ and integer $q \geq 1$,

$$M_{j,q}(t) = \sum_{k \in \mathcal{C}_j} |k|^{2q} \sum_{i=0}^{2} |\widehat{u}_i(t,k)|^2.$$

The zero mode contributes zero to these positive-order moments. From the complete coefficient formula, each nonzero axis mode belongs to $\mathcal{C}_0$ and the two transverse modes belong to $\mathcal{C}_1$. There

are no other nonzero modes. Consequently the *entire* lattice sums are absolutely convergent and exactly

$$M_{0,q} = 4\sum_{n\in\mathbb{Z}} n^{2q} e^{-2(1+\nu t)n^2}, \qquad M_{1,q} = \tfrac{1}{2}e^{-2\nu t}. \tag{14}$$

Gaussian summability proves finiteness; no untracked spectral remainder is discarded. These formulas agree with the source's simpler support labels because both labellings coincide on every nonzero Fourier coefficient. The actual angular transfer is verified in `AngularConeShearCentroid.label_on_spectrum` and `moment_transfer`; full summability and the exact sums are in `GaussianShearSectorMoments`.

Put

$$a = M_{0,1}, \qquad b = M_{1,1}, \qquad d = M_{0,2}, \qquad e = M_{1,2}.$$

For all finite $t \geq 0$ and $\nu \geq 0$ we have

$$0 < b < a < d, \qquad e = b. \tag{15}$$

Indeed $b = \frac{1}{2}e^{-2\nu t} > 0$. The $n = \pm 1$ terms alone give

$$a \geq 8e^{-2(1+\nu t)} = 8e^{-2}e^{-2\nu t} > \tfrac{1}{2}e^{-2\nu t} = b.$$

For the strict constant, $e < 3$ follows for example from the exponential series, so $e^2 < 9 < 16$ and $16e^{-2} > 1$. Every integer satisfies $n^4 \geq n^2$, with strict inequality at $n = 2$. Subtracting the two absolutely convergent sums in (14) therefore gives $d - a > 0$. The equality $e = b$ follows directly from the two unit-frequency transverse modes.

6.3 The specified normalized ratio is strictly above one

Let $T = a + b$, $p_0 = a/T$, $p_1 = b/T$ and $\epsilon_j = p_j - 1/2$. Define the precise ratio being tested by

$$\mathcal{C} = \frac{\epsilon_0(d/a)p_0 + \epsilon_1(e/b)p_1}{((d+e)/T)(\epsilon_0^2 + \epsilon_1^2)}. \tag{16}$$

The formula uses sector enstrophy weights p_j and their own dissipation ratios. It is not a generic definition of every spectral centroid used in the literature. Every denominator is nonzero on this actual flow: $a, b, T, d + e > 0$ and $a > b$ makes $\epsilon_0^2 + \epsilon_1^2 > 0$.

Theorem 6.2 (Reversed angular threshold on one fixed solution)**.** *For the flow* (11) *and this partition,*

$$\mathcal{C} = \frac{(a+b)(d-b)}{(d+b)(a-b)} > 1 \qquad (\nu \geq 0,\ t \geq 0 \text{ finite}).$$

In particular there is no $T_* > 0$ for which $\mathcal{C}(t) < 1$ holds for all $0 < t \leq T_*$ on this fixed flow.

Proof. Use $e = b$ from (15) and $\epsilon_0 = (a-b)/(2T)$, $\epsilon_1 = -(a-b)/(2T)$. The numerator of (16) becomes $(a-b)(d-b)/(2T^2)$ and its denominator is $(d+b)(a-b)^2/(2T^3)$. Cancellation is valid by the established positive denominators and gives the displayed rational expression. Its difference from one is

$$\frac{(a+b)(d-b) - (d+b)(a-b)}{(d+b)(a-b)} = \frac{2b(d-a)}{(d+b)(a-b)} > 0.$$

The last inequality uses the actual strict moment order, not arbitrarily assigned positive numbers. If such an interval existed, evaluation at $t = T_*/2$ would contradict this strict inequality. □

See `GaussianShearCentroid.sourceExpression_eq`, `sourceCentroid_gt_one`, and the actual angular statements `AngularConeShearCentroid.angularCentroid_gt_one` and `no_initial_interval_below_one`.

The example excludes the stated universal threshold test on this actual class and partition. It does not exclude a condition with additional hypotheses, a different angular tessellation or a differently normalized ratio. It also does not produce turbulent evolution or a regularity theorem: its advection is identically zero. Its significance as a calibration is that a fixed classical smooth solution has a genuine infinite tail and reverses the displayed inequality with all complete moments and denominator conditions established.

7 Threshold labels require control of boundary occupation

The smoothness of a field or its sector energies is different from the stability of time spent in a threshold-defined diagnostic region. We record the exact finite-measure distinction. Let (X, μ) be a finite measure space, let $\rho, \zeta : X \to \mathbb{R}$ be measurable scores, and fix thresholds $A, B \in \mathbb{R}$. Define the prioritized label

$$\ell(\rho, \zeta) = \begin{cases} 0, & \rho < A, \\ 1, & \rho \geq A,\ \zeta < B, \\ 2, & \rho \geq A,\ \zeta \geq B. \end{cases}$$

Its occupation totals are $O_j = \mu\{\ell = j\}$. On a finite time interval these are ordinary durations. The relevant boundary is

$$D = \{\rho = A\} \ \cup \ \{\rho \geq A,\ \zeta = B\}. \tag{17}$$

The second threshold is irrelevant where $\rho < A$. In particular $\{\zeta = B\}$ need not be null there. These arbitrary score pairs are not identified with the preceding centroid ratio or a physical evolution law merely by this notation.

Suppose measurable approximations obey $|\rho' - \rho| \leq \varepsilon_\rho$ and $|\zeta' - \zeta| \leq \varepsilon_\zeta$ everywhere, with nonnegative error bounds. Their disagreement set E satisfies

$$E \subseteq \{|\rho - A| \leq \varepsilon_\rho\} \ \cup \ \{\rho \geq A,\ |\zeta - B| \leq \varepsilon_\zeta\}. \tag{18}$$

Indeed, outside the first strip the original and approximating first scores lie on the same side of A: their distance is too small to cross it. If that side is below, both labels are zero. Otherwise $\rho \geq A$, and outside the second strip the two second scores likewise lie on the same side of B. Their labels agree. Each label-region symmetric difference is contained in E; thus

$$\begin{aligned} |O'_j - O_j| &\leq \mu(E) \\ &\leq \mu\{|\rho - A| \leq \varepsilon_\rho\} + \mu\{\rho \geq A, |\zeta - B| \leq \varepsilon_\zeta\}. \end{aligned} \tag{19}$$

The first inequality follows by subtracting the two indicator integrals and bounding the absolute difference by 1_E. For example bounds $C_\rho \varepsilon_\rho^{\gamma_\rho}$ and $C_\zeta \varepsilon_\zeta^{\gamma_\zeta}$ on these actual strip masses immediately give their sum as an occupation error rate. This is an elementary consequence, not a rate asserted without an assumption on the score distribution.

Theorem 7.1. *For the displayed finite-measure classifier, the following are equivalent: $\mu(D) = 0$; every pair of measurable uniform score approximations has disagreement mass tending to zero; and every such approximation has $O'_j \to O_j$ for all three labels. In the last condition it suffices to test labels zero and two for the simultaneous downward shifts $(\rho_n, \zeta_n) = (\rho - \delta_n, \zeta - \delta_n)$, $\delta_n = 1/(n+1)$.*

Proof. Off D, any pointwise-convergent scores eventually have the same label. If $\rho < A$, convergence of the first score alone gives label zero eventually. Otherwise $\rho > A$ and $\zeta \neq B$; the strict distances to both relevant thresholds preserve the label for all sufficiently large n. In fact convergence of the second score is needed only on $\{\rho \geq A\}$. If $\mu(D) = 0$, the disagreement indicators tend to zero almost everywhere and are bounded by the integrable constant one. Dominated convergence gives vanishing disagreement mass, and the first inequality in (19) gives occupation convergence. This proves the forward assertions, even for this weaker almost everywhere score convergence, hence for uniform convergence.

Conversely take the displayed downward shifts, which are uniformly within δ_n of the scores. At a point with $\rho = A$, its old label is one or two and every shifted first score is below A, so the label changes. At a point with $\rho > A, \zeta = B$, the shifted first score remains above A eventually, while the second lies below B: label two changes to one. At points outside D the labels eventually agree by the preceding argument. Thus dominated convergence gives the exact limit

$$\mu\{\ell(\rho_n, \zeta_n) \neq \ell(\rho, \zeta)\} \longrightarrow \mu(D). \tag{20}$$

Universal vanishing disagreement therefore forces $\mu(D) = 0$.

Occupation convergence alone could in principle hide label exchanges, so a further argument is required. Here the downward shifts never increase a label: the prioritized rule is nondecreasing in each coordinate. For ordered labels $j' \leq j$ in $\{0, 1, 2\}$ one has

$$1_{\{j' \neq j\}} \leq (1_{\{j'=0\}} - 1_{\{j=0\}}) + (1_{\{j=2\}} - 1_{\{j'=2\}}).$$

For unchanged labels both sides are zero; for changes $1 \to 0$ and $2 \to 1$ both sides are one; for $2 \to 0$ the right side is two. These are all possible strict decreases. Integrating yields

$$\mu(E_n) \leq (O_{0,n} - O_0) + (O_2 - O_{2,n}).$$

If the two endpoint occupations converge, the nonnegative left side tends to zero. Equation (20) now forces the null-boundary condition. Universal occupation convergence includes this particular shift test, completing every implication. □

The necessity quantifies over all measurable uniform perturbations of these scores. It is not automatically a necessity theorem for a narrower physically attainable family. For the converse direction in such a restricted family, the detecting perturbations would have to be realizable within that family. The following finite-sector example gives a concrete failure without a vanishing denominator or a tie for the largest sector.

8 Smooth sector energies: kinks and full occupation errors

8.1 A maximum preserves a Lipschitz bound, not differentiability

For finitely many functions e_i,

$$|\max_i e_i(t) - \max_i e_i(s)| \leq \max_i |e_i(t) - e_i(s)|.$$

To prove the one-sided inequality, choose an index attaining the maximum at t, replace its value at s by the maximum there, and then reverse s, t. Thus a common Lipschitz constant is retained. It need not retain differentiability even for strictly positive smooth sector energies. Take

$$e_\pm(t) = 1 \pm t + t^2 = (t \pm \tfrac{1}{2})^2 + \tfrac{3}{4}.$$

Their total is $2+2t^2>0$ but their maximum is $1+t^2+|t|$. The normalized maximum concentration is consequently

$$\chi(t)=\frac{\max(e_+(t),e_-(t))}{e_+(t)+e_-(t)}=\frac{1}{2}+\frac{|t|}{2(1+t^2)}.$$

At zero its right derivative is $1/2$ and left derivative is $-1/2$, so it is continuous and locally Lipschitz but not differentiable. This normalized maximum is a different statistic from the squared-weight concentration in the earlier centroid ratio.

There is a useful denominator-controlled bound. Suppose N sector energies are nonnegative at t,s, their totals $E(t),E(s)\geq d>0$, and $|e_i(t)-e_i(s)|\leq K|t-s|$. Write $M=\max_i e_i$. Then $|M(t)-M(s)|\leq K|t-s|$, $|E(t)-E(s)|\leq NK|t-s|$, and $0\leq M(s)/E(s)\leq 1$. Use the exact identity

$$\frac{M(t)}{E(t)}-\frac{M(s)}{E(s)}=\frac{M(t)-M(s)}{E(t)}+\frac{M(s)}{E(s)}\frac{E(s)-E(t)}{E(t)}$$

to obtain

$$|\chi(t)-\chi(s)|\leq\frac{(N+1)K}{d}|t-s|. \tag{21}$$

If $\chi(s)\leq A-\eta$ and the right side is less than η, then $\chi(t)<A$. This is a local threshold-persistence test with actual quantitative hypotheses; a qualitative smoothness claim supplies neither those constants nor an occupation bound.

8.2 A strictly dominant sector can sit at a threshold

On the time interval $[0,1]$ with Lebesgue measure, let $|p|\leq 1/8$ and define constant-in-time energies

$$e_0^p=\tfrac{1}{2}+p,\qquad e_1^p=e_2^p=\tfrac{1}{4}-\tfrac{p}{2}.$$

All are positive, their total is one, and the gap from the first sector to either other sector is $1/4+3p/2\geq 1/16$. Thus the same sector is strictly dominant throughout the family, and its maximum concentration is exactly $\rho_p=1/2+p$. Every energy converges uniformly to the $p=0$ energy as $p\to 0$, with error at most $|p|$. The functions are smooth in time and their parameter. Take the second score $\zeta_p=1$ and thresholds $A=1/2$, $B=2$. For $p=0$ every time has label one, whereas for every $p<0$ every time has label zero. Therefore $O_{0,p}=1$, $O_{0,0}=0$, and their difference is exactly one for arbitrarily small negative p. The relevant boundary has full time measure at $p=0$. The obstruction is threshold mass, despite strict spectral-sector dominance, positive energies and a fixed total. These energies are an actual finite family of scalar diagnostic data; no realization as a Navier–Stokes trajectory is claimed.

The null-boundary and strip arguments are classical continuity-set and dominated-convergence mechanisms. For comparison, the primary margin assumption of Audibert–Tsybakov [5] controls mass near a binary statistical decision threshold. That assumption explicitly excludes equality in its strip and concerns excess risk, not the unweighted disagreement/occupation considered here. In particular positive equality mass cannot be discarded in the present theorem. No statistical minimax rate, learning theorem or new general continuity principle is being asserted.

9 An actual viscous vortex and its axial pressure flux

A further calibration has a genuine stationary Navier–Stokes velocity and its constructed pressure, rather than only an algebraic pressure candidate. It also has an indispensable spatial boundary

limitation. Write $s = x^2 + y^2$, $f = e^{-s}$, and define the regular angular profile and its pressure primitive by

$$g(s) = \frac{1}{2}\int_0^1 e^{-ts}\,dt, \qquad F(s) = \frac{1}{2}\int_0^s g(u)^2\,du. \tag{22}$$

Then $g(0) = 1/2$, and for $s \neq 0$, $g(s) = (1 - e^{-s})/(2s)$. Differentiating the parameter integral any number of times is justified locally uniformly by $|(-t)^k e^{-ts}| \le e^{|s_0|+1}$ for $0 \le t \le 1$ and $|s - s_0| < 1$. Thus g and F are smooth even at the axis, and

$$2g + 2sg' = e^{-s}, \qquad 4g' + 2sg'' = -e^{-s}. \tag{23}$$

The first identity follows by integrating the derivative of te^{-ts} over $[0, 1]$; differentiating it gives the second.

For viscosity $\nu > 0$ and circulation amplitude $\beta \in \mathbb{R}$, set

$$u = (-2\nu x - \beta y g(s),\ -2\nu y + \beta x g(s),\ 4\nu z), \tag{24}$$

$$p = \beta^2 F(s) - 2\nu^2 s - 8\nu^2 z^2. \tag{25}$$

The fluid density is normalized to one.

Theorem 9.1. *The displayed fields are smooth and satisfy pointwise*

$$\nabla \cdot u = 0, \qquad (u \cdot \nabla)u + \nabla p = \nu \Delta u, \qquad \omega := \nabla \times u = (0, 0, \beta e^{-s}).$$

If $\beta \neq 0$, then with $H = \nabla^2 p$ and $B = \nabla \omega$,

$$\operatorname{tr}(B^2) = 0, \qquad \omega \cdot H\omega = -16\nu^2\beta^2 e^{-2s} < 0. \tag{26}$$

For any axial interval $[a, b]$, this nonzero contraction has the exact boundary identity

$$\int_a^b \omega \cdot H\omega\,dz = [\omega_z^2 \partial_z p]_{z=a}^{z=b} = -16\nu^2\beta^2(b - a)e^{-2s}. \tag{27}$$

Proof. The parameter integral proves smoothness. Put

$$v = g(s)(-y, x, 0), \qquad D = \operatorname{diag}(-2\nu, -2\nu, 4\nu),$$

so $u = D(x, y, z) + \beta v$. Direct differentiation gives $\nabla \cdot v = 0$ and $\nabla \times v = (0, 0, 2g + 2sg') = (0, 0, f)$. The linear part has zero trace and zero curl. Moreover

$$(v \cdot \nabla)v = -g^2(x, y, 0), \qquad \Delta v = -2f(-y, x, 0).$$

For the Laplacian, its scalar coefficient is $8g' + 4sg'' = -2f$, by (23). The two cross accelerations satisfy

$$(Dq \cdot \nabla)v + (v \cdot \nabla)Dq = -4\nu(g + sg')(-y, x, 0) = -2\nu f(-y, x, 0).$$

The purely linear acceleration is $(4\nu^2 x, 4\nu^2 y, 16\nu^2 z)$. Finally

$$\nabla p = (\beta^2 x g^2 - 4\nu^2 x,\ \beta^2 y g^2 - 4\nu^2 y,\ -16\nu^2 z),$$

since $F' = g^2/2$. The linear and radial swirl accelerations cancel these pressure terms; the remaining cross acceleration equals $\nu\beta\Delta v$. This proves the actual momentum balance.

The vorticity has only its third component nonzero, is independent of z, and has transverse derivatives $-2\beta x f, -2\beta y f$. Thus B has possible nonzero entries only in its third row and first two columns, so $B^2 = 0$. On the other hand $H_{zz} = -16\nu^2$, which gives the second formula in (26). The axial flux is $\omega_z^2 p_z = -16\nu^2\beta^2 z f^2$. Ordinary integration in z gives (27) and its strict sign when $a < b$. □

There is no cancellation identity obtained by discarding this boundary term. For example, at fixed x, y the unit-slab integral of the full contraction is negative while $\int_0^1 p\,\mathrm{tr}(B^2)dz = 0$. Integrating also over the transverse plane gives the elementary consequence $-8\pi\nu^2\beta^2(b-a)$, since $\int_{\mathbb{R}^2} e^{-2s}dxdy = \pi/2$. This integrated consequence is labelled a human deduction; the source proves the axial identity pointwise in the transverse variables.

The axial velocity $4\nu z$ is unbounded and not periodic. Its square alone has infinite whole-space integral, so the full flow is not a finite-energy three-dimensional example. The transverse vorticity localization does not remove the axial flux. These observations delimit application of the diagnostic to a decaying or periodic global identity. The vortex is classical: Gallay–Maekawa's primary formulas [6], equations (1.3), (1.8)–(1.9), give the background strain and Gaussian circulation profile in their normalization. Here the strain rate is 4ν, the core length is one and the total transverse circulation is $\pi\beta$. No new stationary solution or stability theorem is asserted.

10 A scalar Gaussian observation is not the full Hessian contraction

Let μ be the product Gaussian law with coordinate variance $1/4$. It has density $(2/\pi)e^{-2s}dxdy$. Define

$$r(x,y) = 2s - 1, \qquad \mathcal{R}(P) = \int_{\mathbb{R}^2} rP\,d\mu.$$

Here $\mathbb{E}_\mu r = 0$ and $\|r\|_{L^2(\mu)} = 1$: each coordinate has second moment $1/4$ and fourth moment $3/16$, so $\mathbb{E}s = 1/2$ and $\mathbb{E}s^2 = 1/2$. Consequently $\mathcal{R}$ is a bounded scalar L^2 observation of norm one and ignores constants. It is not the tensor-valued pressure Hessian.

For smooth functions with polynomial growth of the function and its first two derivatives, Gaussian integration by parts gives

$$\mathcal{R}(P) = \frac{1}{8}\mathbb{E}_\mu \Delta P = \frac{1}{2}\mathbb{E}_\mu(xP_x + yP_y). \tag{28}$$

To check the constants, the one-coordinate Gaussian density satisfies $\phi' = -4x\phi$. Integration of its derivative gives $\mathbb{E}P_{xx} = 4\mathbb{E}(xP_x)$, and integration of $(xP\phi)'$ gives $\mathbb{E}[(x^2 - 1/4)P] = \frac{1}{4}\mathbb{E}(xP_x) = \frac{1}{16}\mathbb{E}P_{xx}$. Summing the two coordinate identities and multiplying by two proves (28). Polynomial growth makes every boundary term vanish and all integrals absolutely convergent. The formal source works on a larger explicitly integrable derivative domain, not only polynomials; the constructed pressure below belongs to that domain.

For $s \geq 0$, the defining parameter integral gives $0 < g(s) \leq 1/2$ and $|g'(s)| \leq 1/4$. Thus $0 \leq F(s) \leq s/8$, $\nabla F(s)$ grows at most linearly, and its coordinate second derivatives grow at most quadratically. Indeed $\partial_x F(s) = xg(s)^2$ and $\partial_{xx}F(s) = g(s)^2 + 4x^2 g(s)g'(s)$. These bounds justify the preceding integration by parts for $F(s)$ and for the full pressure (25) at each fixed z.

Proposition 10.1. *Writing $\ell = \log(9/8) > 0$, the exact observations are*

$$\mathcal{R}(F) = \ell/4, \qquad \mathcal{R}(p(\cdot,\cdot,z)) = \beta^2\ell/4 - \nu^2, \tag{29}$$

independently of z. For $\beta = 1$ this scalar observation is zero at $\nu_* = \sqrt{\ell/4}$ and positive at $\nu_*/2$, although the full contraction in (26) *remains strictly negative everywhere at both positive viscosities.*

Proof. Equation (28) gives $\mathcal{R}(F) = \frac{1}{2}\mathbb{E}_\mu[sg(s)^2]$. Squaring the parameter integral in (22) gives

$$g(s)^2 = \frac{1}{4}\int_0^1\int_0^1 e^{-(t+u)s}\,dtdu.$$

The nonnegative integrands permit Tonelli; alternatively $sg(s)^2$ is bounded by $s/4$ and has an integrable Gaussian dominator. Polar integration, whose radial density is $2e^{-2s}ds$, gives

$$\mathbb{E}_\mu[se^{-cs}] = \frac{2}{(2+c)^2} \quad (c \geq 0).$$

Therefore

$$\mathcal{R}(F) = \frac{1}{4}\int_0^1\int_0^1 \frac{dtdu}{(2+t+u)^2}.$$

Integrating first in u yields $(2+t)^{-1}-(3+t)^{-1}$. Integrating in t yields $(\log 3-\log 2)-(\log 4-\log 3) = \log(9/8)$. Also $\mathcal{R}(s) = 2\mathbb{E}s^2 - \mathbb{E}s = 1/2$ and $\mathcal{R}(1) = 0$. Linearity applied to (25) proves the second expression. The assertions at $\nu_*, \nu_*/2$ follow by substitution and $\ell > 0$. The strict tensor sign is already Theorem 9.1 with $\beta = 1$. □

This is an exact mismatch of observables, not a contradiction in the Navier–Stokes pressure. It does not refute a criterion retaining the full Hessian, its proper reference measure or the axial boundary terms.

11 Solenoidal repair and a prescribed base-pressure pairing

A related proposed helical vorticity needs a first-order repair before it can be used even as a divergence-free field. With $f = e^{-s}$, set

$$\widetilde{\omega}_\varepsilon = (\varepsilon f\cos z,\ \varepsilon f\sin z,\ f), \qquad b = x\sin z - y\cos z.$$

Its actual divergence is $-2\varepsilon f(x\cos z + y\sin z)$, nonzero at $(1,0,0)$ for $\varepsilon \neq 0$. The corrected smooth field

$$\omega_\varepsilon = (\varepsilon f\cos z,\ \varepsilon f\sin z,\ f + 2\varepsilon f b) \tag{30}$$

has zero divergence, since the z derivative of its added term is $2\varepsilon f(x\cos z + y\sin z)$. The axial correction is of the same order as the transverse perturbation and cannot be dropped.

Let $B_\varepsilon = \nabla\omega_\varepsilon$. Differentiating the three coordinates gives its first two rows

$$f(-2\varepsilon x\cos z, -2\varepsilon y\cos z, -\varepsilon\sin z), \qquad f(-2\varepsilon x\sin z, -2\varepsilon y\sin z, \varepsilon\cos z),$$

and its third row

$$f(-2x + 2\varepsilon(\sin z - 2xb),\ -2y + 2\varepsilon(-\cos z - 2yb),\ 2\varepsilon(x\cos z + y\sin z)).$$

Multiplying these actual matrices and using $(x\cos z + y\sin z)^2 + b^2 = s$ gives

$$\operatorname{tr}(B_\varepsilon^2) = 4\varepsilon f^2 b + 4\varepsilon^2 f^2(2s-1). \tag{31}$$

In particular its phase integral is $8\pi\varepsilon^2 f^2(2s-1)$, a sign-changing radial quantity. Solenoidality has changed the quadratic trace from that of the uncorrected ansatz; the change is not only the removal of its divergence.

For the next calculation, keep the actual *base* pressure $p_{\nu,\beta}$ from (25), but pair it with the prescribed unit-core field (30). We do not assert that this pressure is induced by a velocity whose curl is ω_ε, or that the latter evolves by Navier–Stokes.

Proposition 11.1. *At every fixed phase z,*

$$\int_{\mathbb{R}^2} p_{\nu,\beta}(x,y,z)\,\mathrm{tr}(B_\varepsilon^2)\,dxdy = \frac{\pi}{2}\varepsilon^2(\beta^2\ell - 4\nu^2). \tag{32}$$

Thus integration on any finite axial interval $[a,b]$ multiplies this value by $b-a$; on $[0,2\pi]$ the result is $\pi^2\varepsilon^2(\beta^2\ell - 4\nu^2)$. These are finite-slab base-pressure pairings, not periodic full-flow identities or full pressure-Hessian contractions.

Proof. The pressure is even under $(x,y) \mapsto (-x,-y)$, while b is odd. Their linear term in (31) integrates to zero. All terms are absolutely integrable, since at fixed z the pressure has at most quadratic transverse growth and the trace is a polynomial in x, y times e^{-2s}. By the physical density of μ, the remaining quadratic contribution equals

$$4\varepsilon^2 \int e^{-2s}(2s-1)p_{\nu,\beta}\,dxdy = 2\pi\varepsilon^2\mathcal{R}(p_{\nu,\beta}).$$

Insert (29) to obtain (32). On a bounded axial interval the same Gaussian polynomial dominator can be chosen uniformly in z, so Fubini and the constant cross-section value justify the axial integral. The quadratic z-dependence of the base pressure does not make the flow periodic merely because this particular observation is independent of z. □

The exact response, solenoidal correction and finite-slab constants are calibration mathematics. The stationary Gaussian vortex and Gaussian integration by parts are classical. Priority for these assembled diagnostics is unresolved; no new vortex-stability theorem, induced helical pressure, finite-energy singularity or unrestricted regularity conclusion is claimed.

12 Formal correspondence and contribution status

Every formal result is frozen at commit `bab688bb6140cd6316c80680095a914ea0d346db`. The family and its small-normalized-stretching quantifiers are in `RestrictedEulerAlignedGrowth`, `RestrictedEulerSpectralDistinction` and `RestrictedEulerArbitraryDepletion`. Actual affine derivatives, momentum and pressure appear in `AffineNavierStokesProfiles` and `AffineStretchingPressureBalance`. The trace test is in `PressureStrainTraceObstruction`. The real periodic derivatives, means and viscous contraction are assembled in `ABCNavierStokesFlow`, `RealBeltramiMeans`, `BeltramiViscousTrace` and `PeriodicPressureSignCounterexample`. The full-spectrum shear and angular ratio use `GaussianInfiniteShear`, `GaussianShearSpectrum`, `GaussianShearSectorMoments`, `GaussianShearCentroid` and `AngularConeShearCentroid`. Finite maximum regularity, relevant threshold boundaries, downward-shift disagreement and sharp endpoint occupation tests use `CorridorMaximumRegularity`, `CorridorOccupation`, `CorridorClassification`, `CorridorBoundarySharpness` and `CorridorOccupationSharpness`. The actual regular Gaussian swirl, viscous strain, axial pressure flux and scalar response use `GaussianSwirlVelocity`, `GaussianSwirlProfileMoments`, `StrainedBurgersVortex`, `BurgersAxialPressureFlux`, `GaussianSwirlLaplace` and `BurgersGaussianPressureResponse`. Solenoidal repair and base-pressure pairings use `HelicalVorticityDiagnostics`, `HelicalRadialGaussianBridge` and `BurgersRepairedTracePairing`. Exact declaration locations, hashes and human proof reorganizations are in the accompanying theorem map. In particular the direct trigonometric proof of $J(U) = 0$ replaces the formal invariant-trace transport proof. Infinite affine energy is the elementary deduction displayed above; the mapped source explicitly proves spatial unboundedness and nonperiodicity, not a named integral theorem.

These examples belong to classical restricted-Euler and Beltrami theory. Cantwell's primary paper was inspected for its model, solution scope and distinction between full-gradient and strain geometry. The primary 2024 pressure-Hessian paper fixes the full/deviatoric conventions; no conflict with its stated equations is alleged. The ABC profile attribution was checked against the primary Xin–Yu–Zlatoš model statement. The examples test the explicit formulas written here, rather than an unspecified published theorem. Their value is reproducible diagnostic mathematics and a formalization companion; novelty or priority for any formulation remains unresolved. The heat-governed shear mechanism is also classical. The primary Gallay–Slijepcevic abstract [4] describes a heat-governed shear regime in a different two-dimensional infinite-cylinder problem. That abstract was inspected for context; none of its asymptotic theorems is used here, and no priority for the displayed two-sector ratio is certified. No Navier–Stokes regularity, finite-energy blow-up or Millennium equivalence claim follows. Complete baseline verification evidence was hash-checked without a redundant full Lean build.

References

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